Vignesh Muralidharan

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PROFESSIONAL SUMMARY

Driven and curious, with a diverse background in Computer Science, Music Production and Design. I possess a strong work ethic and adaptability. I have demonstrated a willingness to learn and grow from every task I take on.
Passionate about creative activities, I am working towards creating new music, games, or even stories.
Currently pursuing my Masters in Game Design and Development at Rochester Institute of Technology.

KEY SKILLS

Programming Languages: Python (Proficient), Java (Intermediate), C# (Intermediate), HTML (Proficient), ReactJS (Working), JavaScript (Working)
Technical Skills: Debugging, API Creation, and Prompt Engineering
Languages Spoken: English (Bilingual), Tamil (Native), Kannada (Limited Working), Sanskrit (Limited Working), German (Limited Working), Chinese (Elementary), Japanese (Elementary)

PROFESSIONAL EXPERIENCE

Intern Apr '24 - Jun '24

Brillio | Bangalore

- *Helped build an Resource Augmented Generation powered internal Artificial Intelligence chatbot to create answers based on company policy and databases*
- *Created a algorithm to parse and format the responses through Prompt Engineering and Microsoft Azure Web Apps creating custom classes and maintaining documentation*
- *Gained skills in python applications such as Langchain, Postman, Semantic Kernel, Resource Augmented Generation and Large Language Model usage*

Intern May '23 - Jun '23

Privafy | Bangalore

- *Built an application to showcase understanding of ReactJS, Flask and PostgreSQL*
- *Created docker containers for each element, and deployed each of them in a MiniKube deployment.*
- *Displayed skills in Go, RestFUL applications and Database access through a Web Application*

EDUCATION

MS in Game Design and Development Present

Rochester Institute of Technology | Rochester, NY

B.Tech in Computer Science and Engineering Jun '21 - May '25

Christ University | Bangalore

With Honors in Artificial Intelligence and Machine Learning

Graduated with GPA 3.61 (4 Point Scale)

PROJECTS

AI-Based Videogame Exploit Investigation Using Evolution-Based Reinforcement Learning Dec '23 - Apr '24

Designed a generational learning algorithm to playtest a game from start to finish, using generational and genetic learning on a population of agents, assigning score based on progress. Allowed the AI to slowly learn from previous generations by utiliizing a natural selection algorithm.

Obtained skills in Game Development, Machine Learning, Genetic Algorithms, Population Study, Natural Selection (in genetic algorithms), Heuristics, Gameplay Agents

Custom Dice Roller using OpenGL Jul '23 - Nov '23

Created an OpenGL application to generate and simulate in 3D various polyhedral dice: Tetrahedron (D4), Cube (D6), Octahedron (D8), Pentagonal Trapezohedron (D10), Dodecahedron (D12), and Icosahedron (D20). Implemented textures with user selection, with each die rolling values between 1 and n inclusive.

Obtained skills in OpenGL, 3D Modeling, Graphics Programming, UI Design, Animation Techniques, Shader Programming, and Texture Design